

Exam Date & Time: 27-Nov-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VII Semester - End Semester Examination - Nov / Dec -2024

Cloud Computing [IT-4045]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) A company "Manipal Infosystems" is considering migrating their IT infrastructure to the cloud. What are the main types of cloud services they should consider for migration? Explain. (5)
 - A)
 - B) A startup is looking to launch a new web application quickly and cost-effectively. How can PaaS help them achieve this goal? (3)
 - C) In a CIA Triad what do you mean by Integrity? (2)
- 2) In a cloud computing environment, four users (User D, User E, User F, and User G) are submitting tasks to a shared resource pool. The system uses a fair queueing algorithm to allocate resources. Each user submits tasks at different rates and requires different amounts of resources.
 - A)
 - **User D** submits tasks at a rate of 6 tasks per minute, each requiring 1.5 units of resource.
 - **User E** submits tasks at a rate of 4 tasks per minute, each requiring 2 units of resource.
 - **User F** submits tasks at a rate of 5 tasks per minute, each requiring 1 unit of resource.
 - **User G** submits tasks at a rate of 3 tasks per minute, each requiring 2.5 units of resource.(5)

The total available resource in the system is 25 units per minute. Based on the above scenario the following questions:

 1. Calculate the total resource demand per minute for each user.
 2. Determine the fair share of resources each user should receive per minute based on their demand and the total available resources.
 3. Calculate the number of tasks each user can complete per minute with their fair share of resources.
 4. Evaluate the system's efficiency by comparing the total resource utilization with the available resources.

- B) Identify the components of VDC Network Infrastructure and explain its functionalities. (3)
- C) State how VM Theft happens in the cloud environment. (2)
- 3) A data centre company called XYZ wants to virtualize its networks. Analyse how network virtualization can be implemented to service its customers in detail. (5)
- A)
- B) Mention and explain the steps used to protect the network in virtualized environment. (3)
- C) How do you enforce security at compute level? Give a detailed procedure to be followed. (2)
- 4) Consider for a data set containing Name and Average_Runs key value pair, explain with a figure how would you proceed using Map-Reduce framework to find maximum scored player name. (5)
- A)
- B) Identify the benefits of VLAN Tagging. How it works? Explain in detail. (3)
- C) Manipal Tech want to deploy cloud environment in their office using cloudsimsim tool. Analyse the steps required to simulate a cloud environment in detail. (2)
- 5) An Educational training based MNC aims to modernize its IT infrastructure and enhance its digital services. How it will adapt its Cloud Infrastructure Framework for its enhanced operations? Provide and explain with the suitable diagram. (5)
- A)
- B) Identify and explain the steps involved using RAID 5 - striped parity in storage protection method. (3)
- C) What are the parameters need to be specified when creating a cloudlet in CloudSim? (2)

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